BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2021

CHEMISTRY

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Chemistry (Lahore, Tenth 2021) (Paper - II) (Objective Type) (Group - I) (Session 2017-2019 to 2019-2021)

Time Allowed: 15 Min. | **Maximum Marks:** 12

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. Glucose is :

(A) Hexahydroxy aldehyde
(B) Hexahydroxy ketone
(C) Pentahydroxy aldehyde
(D) Pentahydroxy ketone

Q2. Matte is mixture of:

(A) FeS and CuS
(B) Cu_2O and FeO
(C) Cu_2S and FeS
(D) Cu_2S_2 and FeS

Q3. The functional group -COOH is found in :

(A) Carboxylic acid
(B) Aldehyde
(C) Alcohols
(D) Esters

Q4. Which gas protects the earth surface from ultraviolet radiations :

(A) CO_2
(B) CO
(C) N_2
(D) O_3

Q5. Formula of urea is :

(A) NH_2COONH_4
(B) NH_2COONH_2
(C) NH_2CONH_4
(D) NH_2CONH_2

Q6. For a reaction between PCl_3 and Cl_2 to form PCl_5 the unit of Kc are:

(A) mol/dm^3 (B) $mol^{-1}dm^{-3}$

(C) $mol^{-1}dm^3$ (D) $mol\ dm^3$

Q7. Alkenes are prepared from alcohols by process called :

(A) Dehydrogenation (B) Dehalogenation

(C) Dehydration (D) Dehydrohalogenation

Q8. The new substance formed in chemical reaction is :

(A) Reactant (B) Product

(C) Forward (D) Reverse

Q9. Just above the earth's surface is :

(A) Mesosphere (B) Stratosphere

(C) Thermosphere (D) Troposphere

Q10. The reduction of alkyl halides takes place in the presence of:

(A) Zn/HCl (B) Na/HCl

(C) Mg/HCl (D) Cu/HCl

Q11. Which ion is the Conjugate base of sulphuric acid :

(A) SO_3^{2-} (B) S^{2-}

(C) HSO_3^- (D) HSO_4^-

Q12. Water dissolves non-ionic compound by :

(A) Ion ion forces (B) Ion dipole forces

(C) Dipole dipole forces (D) Hydrogen bonding

Chemistry (Lahore, Tenth 2021) (Paper - II) (Essay Type) (Group - I) (Session 2017-2019 to 2019-2021)

Time Allowed: 1.45 hrs | **Maximum Marks:** 48

QPART - I.

Q2. (Marks 10) Write short answers to any FIVE (5) questions:

i Write any two macroscopic properties of reverse reaction.

ii Write the equilibrium constant expression for the following reaction : $2NO_2(g) \rightleftharpoons N_2O_4(g)$

iii Write two uses of aluminium hydroxide.

iv Define chemical equilibrium state.

v What are limitations of Arrhenius concept?

vi Write the name and formula of any two naturally occurring acids.

vii How is carboxylic group tested?

viii Write two uses of acetic acid.

Q3. (Marks 10) Write short answers to any FIVE (5) questions:

i Give two sources of alkanes.

ii Give two physical properties of alkynes.

iii What do you know about hydrogenation of alkenes?

iv Give general formula of triglyceride.

v Give name and formula of a fatty acid.

vi Define green house effect.

vii Give two effects of acid rain.

viii How acid rain is produced?

Q4. (Marks 10) Write short answers to any FIVE (5) questions:

i Water is a universal solvent. Give an example.

ii What are soft and hard water?

iii How detergents make the water unfit for aquatic life?

iv Write a brief note on cholera.

v Define : (i) Gangue (ii) Metallurgy.

vi What is gravity separation?

vii Give raw materials for preparation of urea.

viii How roasting is carried out?

QPART - II. Note: Attempt any TWO questions.

Q5.(a). (Marks 4) Write down five methods for the preparation of salts.

Q5.(b). (Marks 5) Write a note on halogenation of alkanes.

Q6.(a). (Marks 4) Write down the characteristics of homologous series.

Q6.(b). (Marks 5) Write four properties of monosaccharides.

Q7.(a). (Marks 4) Where does ozone layer lie in atmosphere? How it is depleting and how can we prevent its depletion?

Q7.(b). (Marks 5) How polarity of water molecule plays its role to dissolve the substances?

GROUP – II

Chemistry (Lahore, Tenth 2021) (Paper - II) (Objective Type) (Group - II) (Session 2017-2019 to 2019-2021)

Time Allowed: 15 Min. | **Maximum Marks:** 12

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. The pH of acid rain due to air pollutants become :

(A) 3 (B) 4

(C) 8 (D) 9

Q2. Substitution reaction is the characteristic of:

(A) Alkanes (B) Alkenes

(C) Alkyl (D) None of these

Q3. Concentration is a :

(A) Mixing technique (B) Boiling technique

(C) Separating technique (D) Cooling technique

Q4. For the reaction $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ what will be present in the equilibrium mixture.

(A) NH_3 only (B) N_2, H_2 and NH_3

(C) N_2 and H_2 only (D) H_2 only

Q5. Which one of the following is a raw material for urea :

(A) CO (B) CO_2

(C) N_2 (D) NO_2

Q6. The colour of HI is :

(A) Colourless (B) Purple

(C) Green (D) Red

Q7. Which compound causes temporary hardness of water:

(A) $CaSO_3$ (B) $MgSO_4$

(C) $MgCl_2$ (D) $Ca(HCO_3)_2$

Q8. The formula of ethyl radical is :

(A) $C_3H_7–$ (B) $C_2H_5–$

(C) $C_2H_4–$ (D) $C_3H_6–$

Q9. Just above the earth's surface is :

(A) Mesosphere (B) Stratosphere

(C) Thermosphere (D) Troposphere

Q10. Which one of the following is tasteless.

(A) Starch (B) Glucose

(C) Fructose (D) Sucrose

Q11. The formula of butyne is :

(A) C_3H_4 (B) C_4H_6

(C) C_4H_{10} (D) C_4H_8

Q12. Which one of the following is primary pollutant:

(A) HNO_3 (B) H_2SO_4

(C) CO (D) H_2CO_3

Chemistry (Lahore, Tenth 2021) (Paper - II) (Essay Type) (Group - II) (Session 2017-2019 to 2019-2021)

Time Allowed: 1.45 hrs | **Maximum Marks:** 48

QPART - I.

Q2. (Marks 10) Write short answers to any FIVE (5) questions:

i What is meant by static equilibrium?

ii Give any two characteristics of reversible reaction.

iii Define reversible reaction and give example.

iv Define acid and base according to Arrhenius concept.

v Write two uses of acetic acid.

vi Define pH. What is the pH of pure water?

vii Define functional group with an example.

viii What is the difference between n-propyl and iso-propyl radicals. Explain with structure.

Q3. (Marks 10) Write short answers to any FIVE (5) questions:

i What do you know about hydrogenation of alkenes? Give an example.

ii Write any two uses of acetylene.

iii Write dehalogenation of tetrahalides.

iv How plants synthesize carbohydrates?

v Define lipids also write general formula of triglycerides.

vi Differentiate between pollutants and contaminants.

vii Write chemical reaction occurring in the mid-stratosphere.

viii What is Ozone hole?

Q4. (Marks 10) Write short answers to any FIVE (5) questions:

i Describe the two effects of using polluted water.

ii What is the jaundice?

iii Which diseases are caused by acute cadmium poisoning?

iv How decaying plants consume oxygen.

v Define ores.

vi What is smelting?

vii What is electromagnetic separation?

viii Write the name of two urea manufacturing units.

QPART - II. Note: Attempt any TWO questions.

Q5.(a). (Marks 4) Write any five uses of bases.

Q5.(b). (Marks 5) Write two methods of preparation of alkynes.

Q6.(a). (Marks 4) How alkyl radicals are formed? Give name and formulae of any four alkyl radicals.

Q6.(b). (Marks 5) Explain that amino acids are building blocks of proteins.

Q7.(a). (Marks 4) Give the characteristics of troposphere. Why temperature decreases upward in this sphere?

Q7.(b). (Marks 5) How polarity of water molecule plays its role to dissolve the substances?